Math 105L **Lab 5 Part 1**

Alternative Derivative Definition & Discovery of the Power Rule

1. The Alternative Limit Definition of the Derivative

Recall the standard definition of the derivative at a point x:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

By performing the substitution $\mathbf{z = x + h}$:

- Replacing $x + h$ gives $f(z)$.
- Since $z = x + h$, solving for h yields $\mathbf{h = z - x}$.
- As $h \to 0$, $x + h \to x$, which means $\mathbf{z \to x}$.

Substituting these three pieces yields the **alternative limit definition of the derivative**:

$$\mathbf{f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}}$$

2. Demonstration: Verifying $f(x) = x^2 \implies f'(x) = 2x$

Using the alternative limit definition on $f(x) = x^2$:

$$f'(x) = \lim_{z \to x} \frac{z^2 - x^2}{z - x} = \lim_{z \to x} \frac{(z-x)(z+x)}{z - x} = \lim_{z \to x} (z + x) = x + x = \mathbf{2x} \qquad \checkmark$$

3. Factorization Patterns of $z^n - x^n$

To differentiate general monomials $f(x) = x^n$, we factor $z^n - x^n$ by extracting $(z - x)$:

$$\begin{aligned}
z^2 - x^2 &= (z-x)(\mathbf{z + x}) \\
z^3 - x^3 &= (z-x)(\mathbf{z^2 + zx + x^2}) \\
z^4 - x^4 &= (z-x)(\mathbf{z^3 + z^2x + zx^2 + x^3}) \\
z^5 - x^5 &= (z-x)(\mathbf{z^4 + z^3x + z^2x^2 + zx^3 + x^4}) \\
&\vdots \\
z^n - x^n &= (z-x)\big(\mathbf{z^{n-1} + z^{n-2}x + z^{n-3}x^2 + \cdots + zx^{n-2} + x^{n-1}}\big)
\end{aligned}$$

4. Rigorous Proof of the Power Rule for Positive Integers n

Applying the alternative limit definition to $f(x) = x^n$:

$$\begin{aligned}
f'(x) &= \lim_{z \to x} \frac{z^n - x^n}{z - x} \\
&= \lim_{z \to x} \frac{(z-x)\big(z^{n-1} + z^{n-2}x + \cdots + zx^{n-2} + x^{n-1}\big)}{z - x} \\
&= \lim_{z \to x} \big(z^{n-1} + z^{n-2}x + \cdots + zx^{n-2} + x^{n-1}\big)
\end{aligned}$$

Counting the terms:

- The polynomial factor contains exactly $\mathbf{n}$ terms.
- As $z \to x$, each individual term becomes:

$$z^{n-1} \to x^{n-1}, \quad z^{n-2}x \to x^{n-2}x = x^{n-1}, \quad \ldots, \quad x^{n-1} \to x^{n-1}$$

- Summing these n identical terms together yields:

$$\mathbf{f'(x) = \underbrace{x^{n-1} + x^{n-1} + \cdots + x^{n-1}}_{\mathbf{n}\text{ terms}} = nx^{n-1}} \qquad \blacksquare$$